

Lab 3: Architectural Style Selection – Public Bus Live Tracking & Crowding Estimator

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Architecture Selection

We chose **Microservices Architecture** for the Public Bus Live Tracking & Crowding Estimator System.

Architectural Choice

The system is divided into independent services such as GPS Data Injection, Bus Tracking, ETA Estimation, Crowding Estimation, Commuter Information, and Fleet Control. Each service performs a specific function and communicates with other services through defined interfaces.

Reason 1 – Scalability

The system must handle GPS telemetry from up to 1,000 active buses simultaneously. Microservices allow individual services such as the Bus Tracking and GPS Data Injection services to be scaled independently according to their workload.

Reason 2 – Independent Services

ETA estimation and crowding estimation are separate real-time functions. Using microservices allows these services to be developed, maintained, and deployed independently without affecting the entire system.

Security Advantage

Microservices can use authenticated and authorized communication between services. Access to sensitive transit and passenger-related information can therefore be restricted to authorized users and services.

Performance Benefit

Real-time services such as GPS processing, ETA calculation, and crowding estimation can be independently scaled during peak traffic conditions. This helps the system process incoming data efficiently and provide timely information to commuters and fleet controllers.